

Lec 1

Neural Network (introduction)

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Introduction

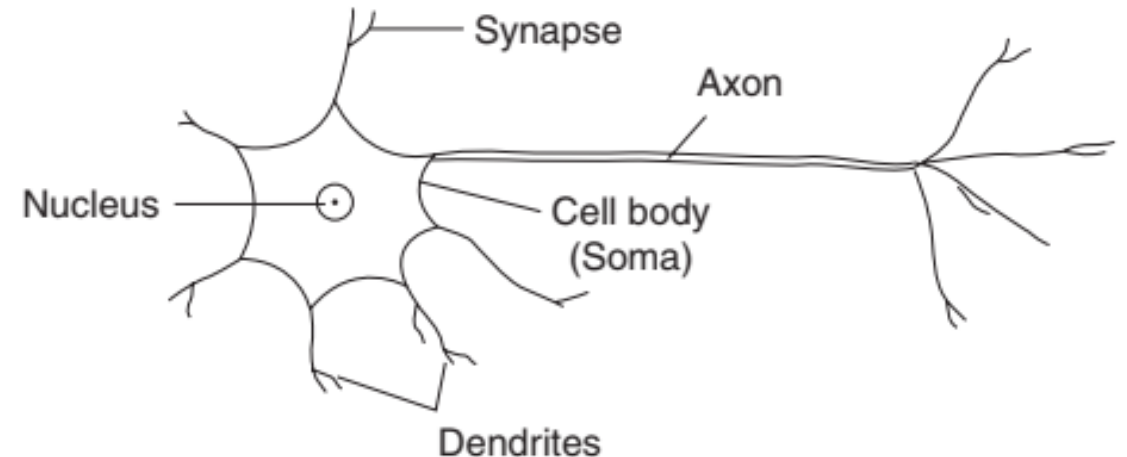
- A neural network is an algorithm, whose design was inspired by the design and functioning of animal brains and components thereof.
- The computing world has a lot to gain from neural networks, also known as artificial neural networks or neural net.
- The neural networks have the ability to learn by example, which makes them very flexible and powerful.
- The human brain is an amazing processor. The most basic element of the human brain is a specific type of cell, known as neuron.
- Because neurons aren't slowly replaced, it is assumed that they provide us with our abilities to remember, think and apply previous experiences.
- The human brain comprises about 100 billion neurons. Each neuron can connect with up to 200,000 other neurons,

Biological Neural Network

- It is well-known that the human brain consists of a huge number of neurons, with numerous interconnections. A schematic diagram of a biological neuron.
- The biological neuron consists of three main parts:
 - 1. Soma or cell body – where the cell nucleus is located.
 - 2. Dendrites – where the nerve is connected to the cell body.
 - 3. Axon – which carries the impulses of the neuron.

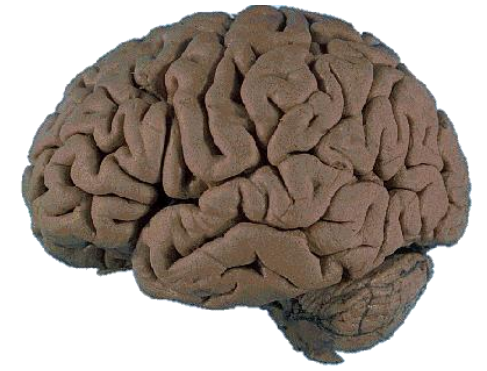
The dendrites collect stimuli from the neighbouring neurons and pass it on to soma which is the main body of the cell.

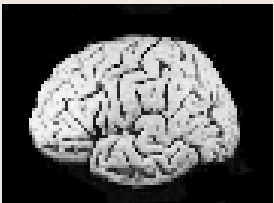
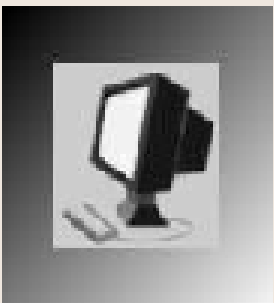
The soma accumulates the stimuli received through the dendrites. It 'fires' when sufficient stimuli is obtained. When a neuron fires it transmits its own stimulus through the axon.

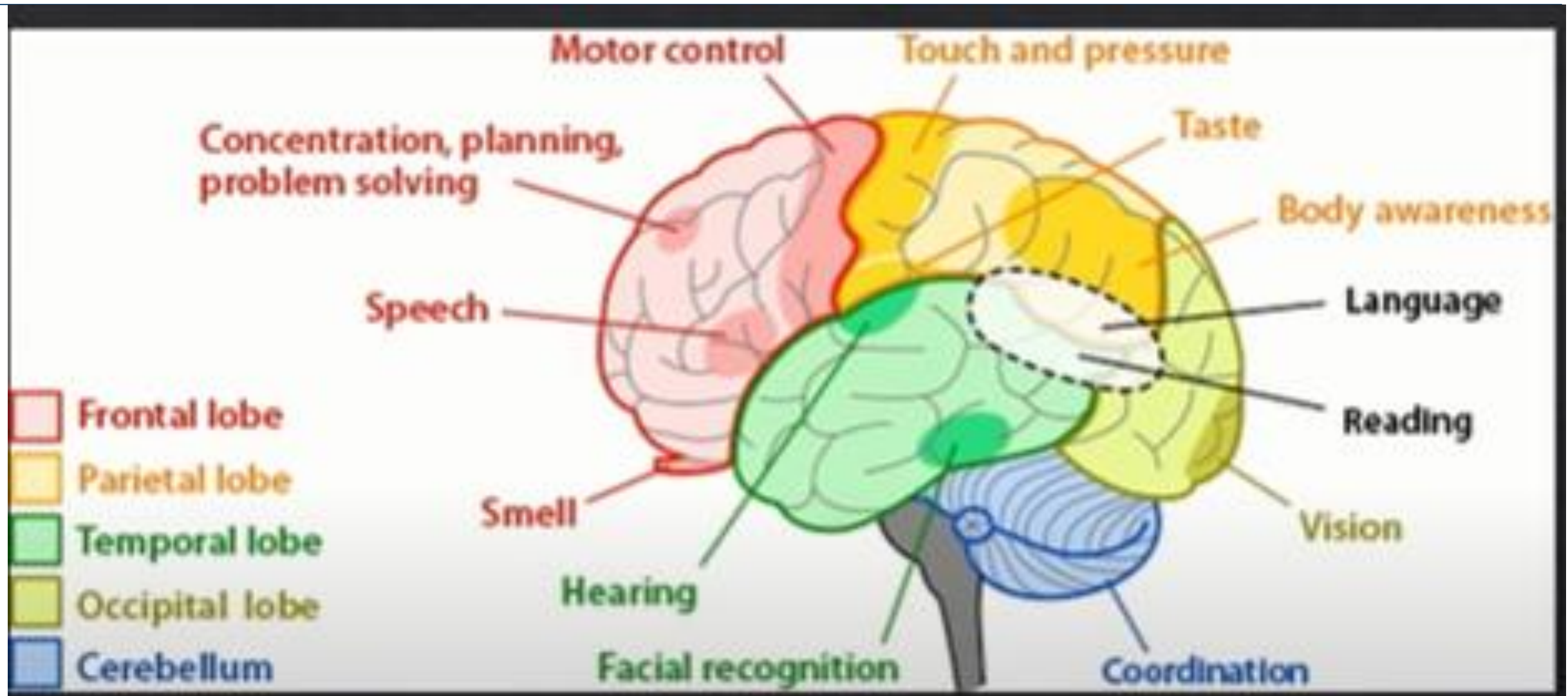


BRAIN COMPUTATION

The **human brain** contains about 10 billion nerve cells, or neurons. On average, each neuron is connected to other neurons through approximately 10,000 synapses.

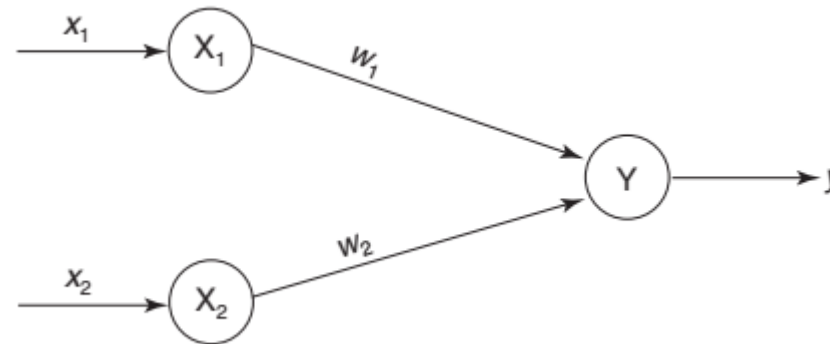


	processing elements	element size	energy use	processing speed	style of computation	fault tolerant	learns	intelligent, conscious
	10^{14} synapses	10^{-6} m	30 W	100 Hz	parallel, distributed	yes	yes	usually
	10^8 transistors	10^{-6} m	30 W (CPU)	10^9 Hz	serial, centralized	no	a little	not (yet)



- An artificial neural network (ANN) may be defined as an **information-processing model** that is **inspired** by the way biological nervous systems, such as the brain, process information.
- This model tries to replicate only the **most basic functions of the brain..**
- ANN is **composed** of a large number of highly interconnected processing elements (neurons) working in unison to solve specific problems.
- **Artificial neural networks**, like people, **learn by example**.
- An ANN is configured for a specific application, such as pattern recognition or data classification through a learning process.

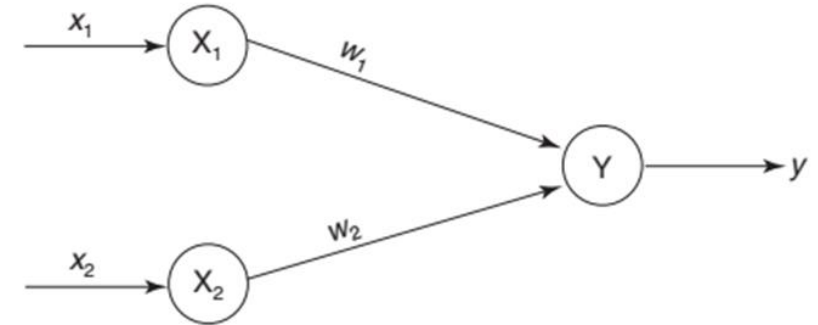
- Each neuron is connected with the other by a **connection link**.
- Each connection link is **associated with weights** which contain information about the input signal.
- This information is used by the neuron net to solve a particular problem.



- It should be noted that each neuron has an **internal state** of its own. This internal state is **called the activation** or activity level of neuron, which is the function of the inputs the neuron receives.
- The activation signal of a neuron is transmitted to other neurons. Remember, a neuron can send only one signal at a time, which can be transmitted to several other neurons.
- To depict the basic operation of a neural net, consider a set of neurons, say X1 and X2, transmitting signals to another neuron, Y. Here X1 and X2 are input neurons, which transmit signals, and Y is the output neuron, which receives signals.

- Input neurons X1 and X2 are connected to the output neuron Y, over a weighted interconnection links (W1 and W2)
- For the above simple neuron net architecture, the net input has to be calculated in the following way:

$$y_{in} = x_1 w_1 + x_2 w_2$$

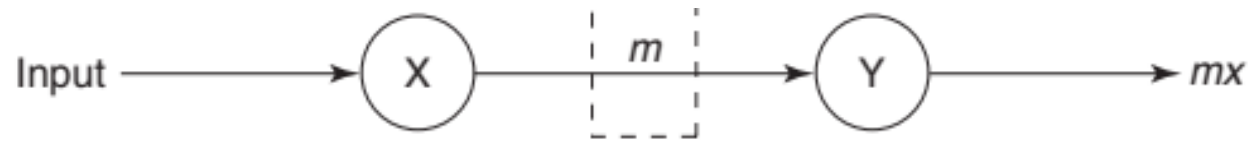


where x1 and x2 are the activations of the input neurons X1 and X2, i.e., the output of input signals. The output y of the output neuron Y can be obtained by applying activations over the net input, i.e., the function of the net input:

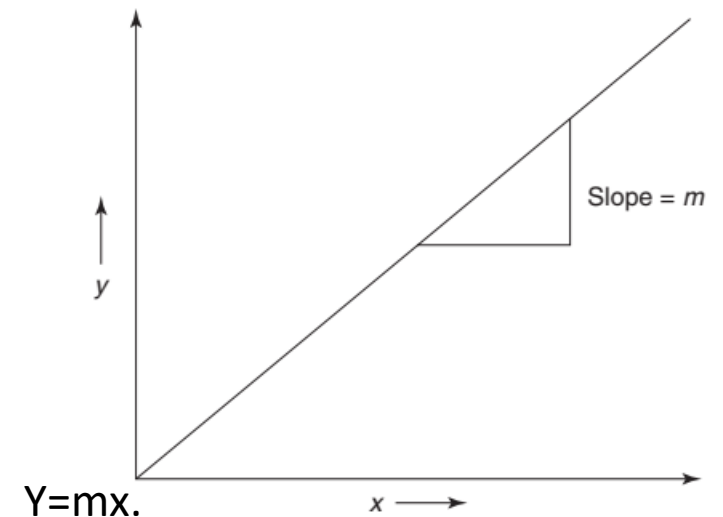
$$y = f(y_{in})$$

Output = Function (net input calculated)

- The function to be applied over the net input is called **activation function**.
- There are various activation functions, which will be discussed in the forthcoming sections. The above calculation of the net input is similar to the calculation of output of a pure linear straight line equation ($y = mx$).



Here, to obtain the output y , the slope m is directly multiplied with the input signal. This is a linear equation. Thus, when slope and input are linearly varied, the output is also linearly varied,

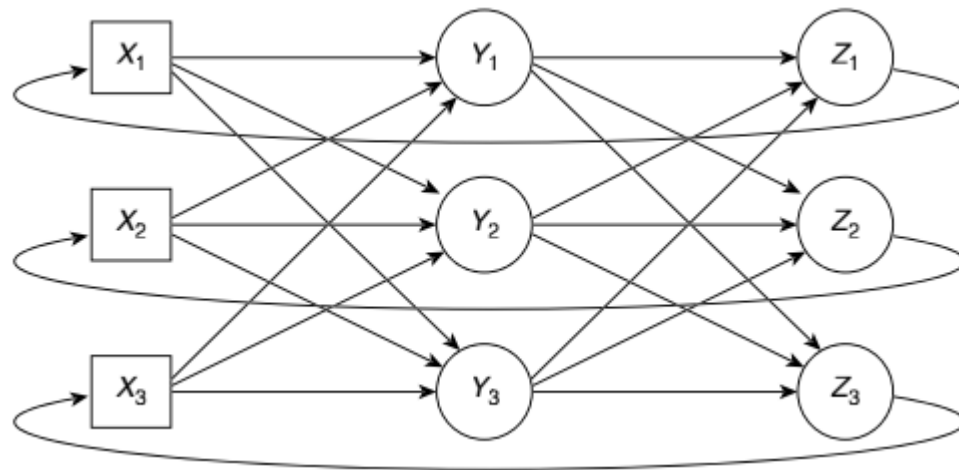
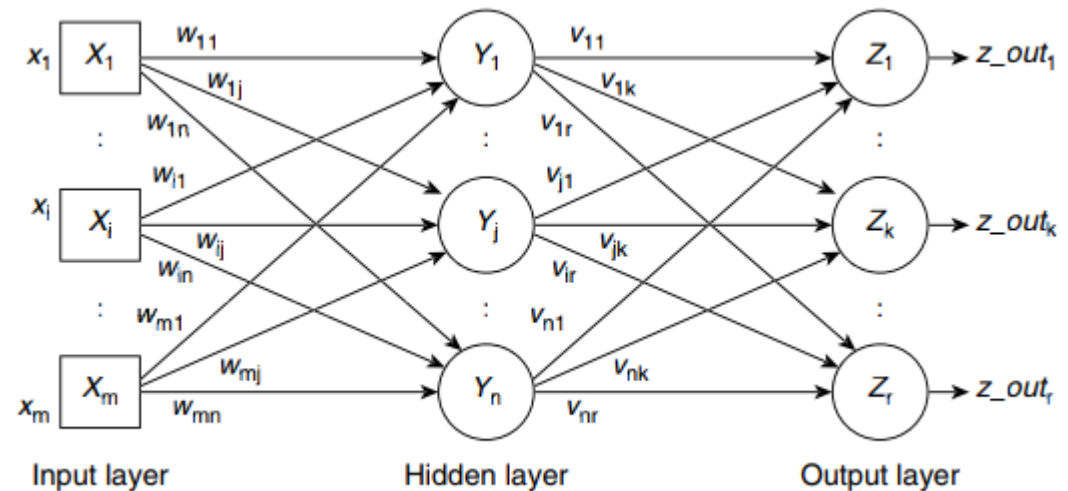
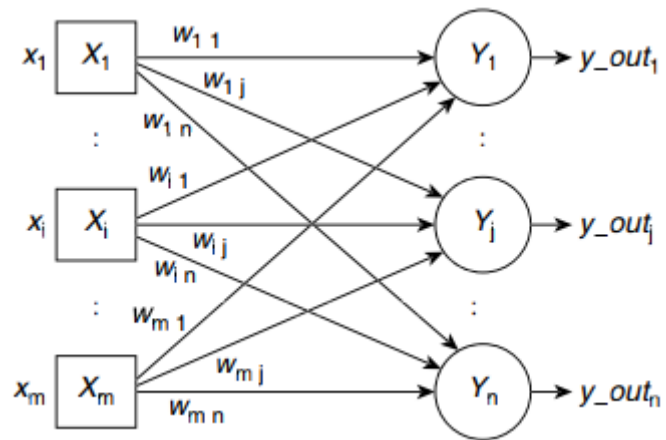


BASIC MODELS OF ARTIFICIAL NEURAL NETWORK

The models of ANN are specified by the three basic entities namely:

- 1. the model's synaptic interconnections (connections)
- 2. the training or learning rules adopted for updating and adjusting the connection weights
- 3. their activation functions

- The arrangement of neurons to form layers and the connection pattern formed within and between layers is called the network architecture. There exist five basic types of neuron connection architectures. They are:
 - 1. single-layer feed-forward network;
 - 2. multilayer feed-forward network;
 - 3. single node with its own feedback;
 - 4. single-layer recurrent network;
 - 5. multilayer recurrent network.

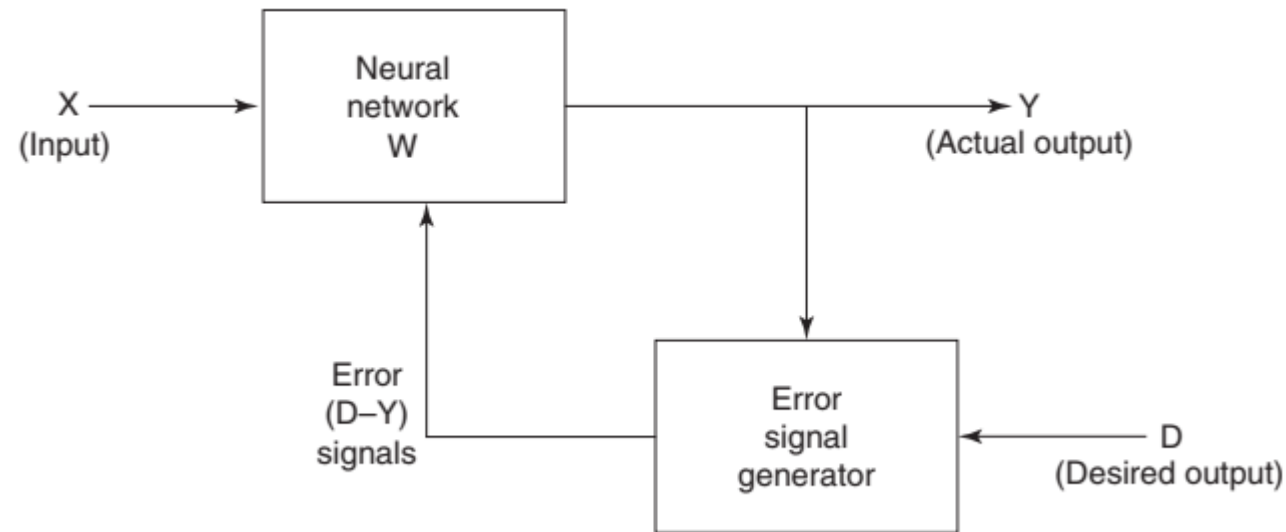


Learning

- The main property of an ANN is its **capability to learn**. Learning or training is a process by means of which a neural network adapts itself to a stimulus **by making proper parameter adjustments**, resulting in the production of desired response.
- Broadly, there are two kinds of learning in ANNs:
 1. Parameter learning: It **updates the connecting weights** in a neural net.
 2. Structure learning: It focuses on the change in **network structure** (which includes the number of processing elements as well as their connection types).
- the learning in an ANN can be generally classified into three categories as: **supervised learning; unsupervised learning; reinforcement learning**.

Supervised Learning

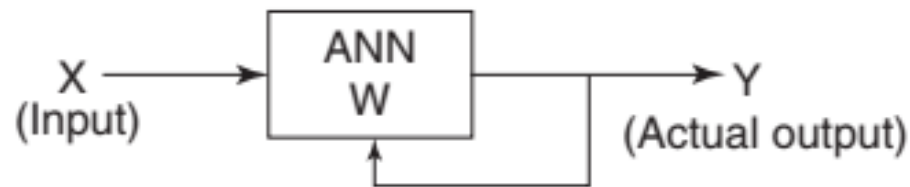
- The learning here is performed with the help of a teacher.
- each input vector requires a corresponding target vector, which represents the desired output. The input vector along with the target vector is called training pair. The network here is informed precisely about what should be emitted as output.



- During training, the input vector is presented to the network, which results in an output vector.
- This output vector is the actual output vector.
- Then the actual output vector is compared with the desired (target) output vector.
- If there exists a difference between the two output vectors then an error signal is generated by the network.
- This error signal is used for adjustment of weights until the actual output matches the desired (target) output.

Unsupervised Learning

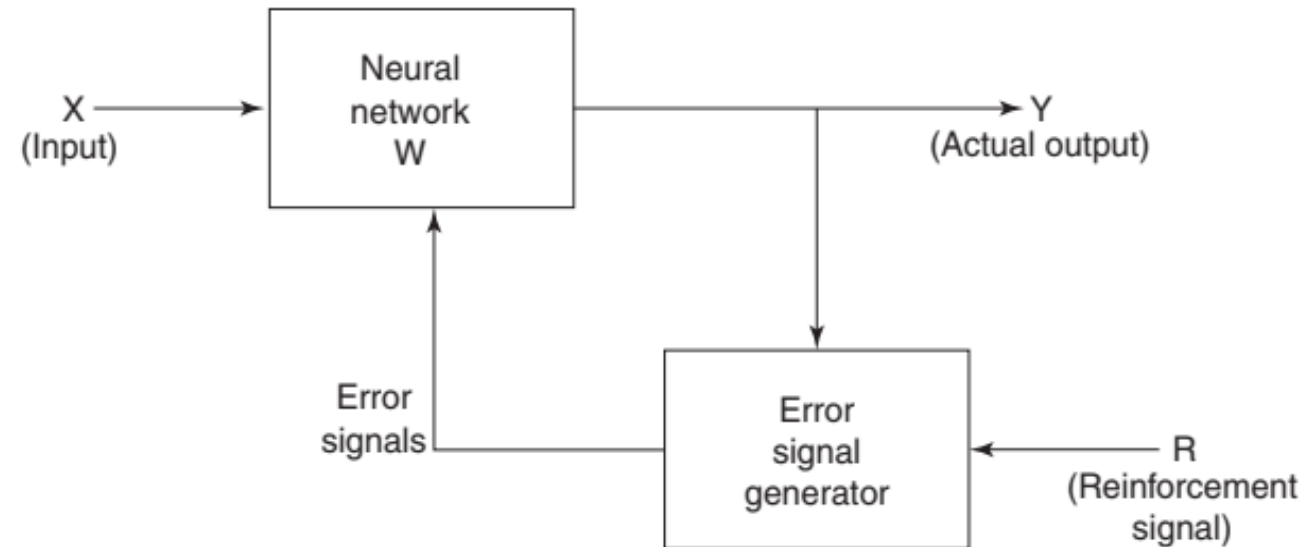
- In ANNs following unsupervised learning, the input vectors of similar type are grouped without the use of training data to specify how a member of each group looks or to which group a number belongs.
- In the training process, the network receives the input patterns and organizes these patterns to form clusters. When a new input pattern is applied, the neural network gives an output response indicating the class to which the input pattern belongs. If for an input, a pattern class cannot be found then a new class is generated.



- it is clear that there is no feedback from the environment to inform what the outputs should be or whether the outputs are correct.
- In this case, the network must itself discover patterns, regularities, features or categories from the input data and relations for the input data over the output.
- While discovering all these features, the network undergoes change in its parameters. This process is called self-organizing in which exact clusters will be formed by discovering similarities and dissimilarities among the objects.

Reinforcement learning

- This learning process is similar to supervised learning. In the case of supervised learning, the correct target output values are known for each input pattern. But, in some cases, less information might be available.
- For example, the network might be told that its actual output is only “50% correct” or so. Thus, here only critic information is available, not the exact information. The learning based on this critic information is called reinforcement learning and the feedback sent is called reinforcement signal.



- The reinforcement learning is a form of supervised learning because the network receives some feedback from its environment.
- However, the feedback obtained here is only evaluative and not instructive.
- The external reinforcement signals are processed in the critic signal generator, and the obtained critic signals are sent to the ANN for adjustment of weights properly so as to get better critic feedback in future. The reinforcement learning is also called learning with a critic as opposed to learning with a teacher, which indicates supervised learning. So, now you've a fair understanding of the three generalized learning rules used in the training process of ANNs.

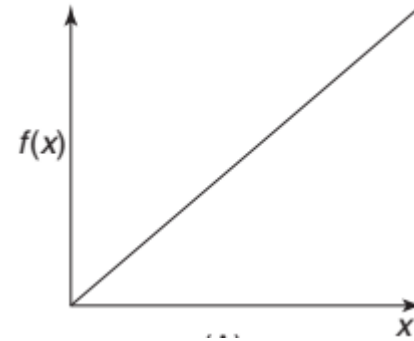
Activation Functions

- The activation function is applied over the net input to calculate the output of an ANN.
- The information processing of a processing element can be viewed as consisting of two major parts: input and output. An integration function (say f) is associated with the input of a processing element.
- This function serves to combine activation, information or evidence from an external source or other processing elements into a net input to the processing element. The nonlinear activation function is used to ensure that a neuron's response is bounded – that is, the actual response of the neuron is conditioned or dampened as a result of large or small activating stimuli and is thus controllable.

- There are several activation functions. Let us discuss a few in this section:

- 1. **Identity function**: It is a linear function and can be defined as

$$f(x) = x \quad \text{for all } x$$

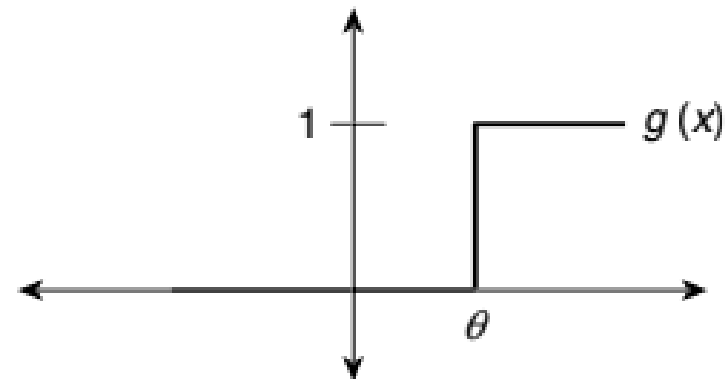


- The output here remains the same as input. The input layer uses the identity activation function.

- 2. **Binary step function**: This function can be defined as

$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ 0 & \text{if } x < \theta \end{cases}$$

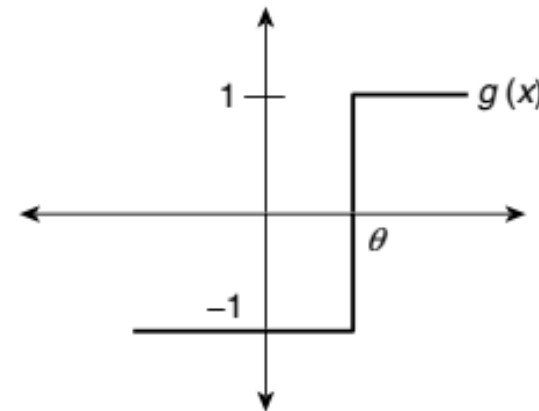
- where θ represents the threshold value. This function is most widely used in single-layer nets to convert the net input to an output that is a binary (1 or 0).



- **3. Bipolar step function:** This function can be defined as

$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ -1 & \text{if } x < \theta \end{cases}$$

- where θ represents the threshold value. This function is also used in single-layer nets to convert the net input to an output that is bipolar (+1 or -1).



- 4. **Sigmoidal functions**: The sigmoidal functions are widely used in back-propagation nets because of the relationship between the value of the functions at a point and the value of the derivative at that point which reduces the computational burden during training. Sigmoidal functions are of two types:

- Binary sigmoid function: It is also termed as logistic sigmoid function or unipolar sigmoid function. It can be defined as

$$f(x) = \frac{1}{1 + e^{-\lambda x}}$$

where λ is the steepness parameter. The derivative of this function is

Here the range of the sigmoid function is from 0 to 1.

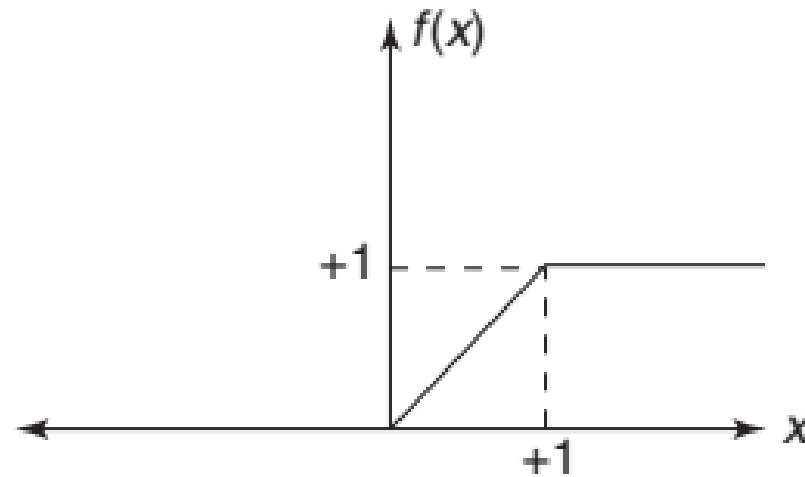
- **Bipolar sigmoid function:** This function is defined as

$$f(x) = \frac{2}{1 + e^{-\lambda x}} - 1 = \frac{1 - e^{-\lambda x}}{1 + e^{-\lambda x}}$$

where λ is the steepness parameter and the sigmoid function range is between -1 and +1.

- 5. **Ramp function**: The ramp function is defined as

$$f(x) = \begin{cases} 1 & \text{if } x > 1 \\ x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{if } x < 0 \end{cases}$$



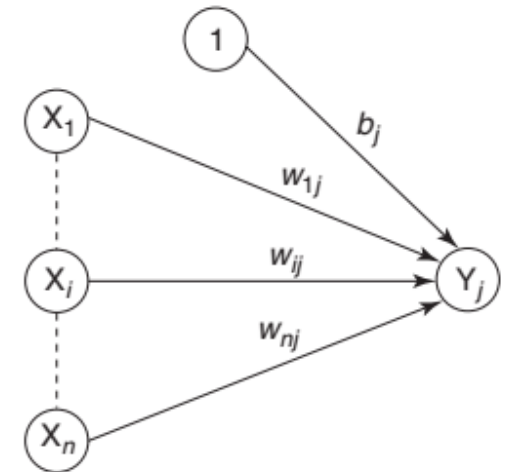
- **Weights**

- Weight is a parameter which contains information about the input signal. This information is used by the net to solve a problem.
- In ANN architecture, every neuron is connected to other neurons by means of a directed communication link and every link is associated with weights.
- W_{ij} is the weight from processing element 'i' source node to processing element 'j' destination node.
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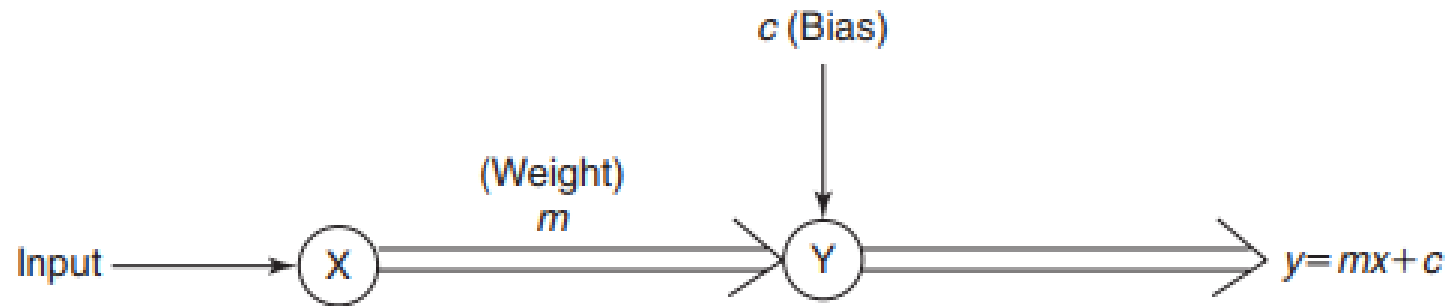
- **Bias (b)**

- The bias is a constant value included in the network. Its impact is seen in calculating the net input. The bias is included by adding a component $x_0 = 1$ to the input vector X .
- Bias can be positive or negative. The positive bias helps in increasing the net input of the network. The negative bias helps in decreasing the net input of the network.

$$y_{inj} = b_j + \sum_{i=1}^n x_i w_{ij}$$



- Block diagram for straight line.



- **Threshold (Θ)**

- Threshold is a set value used in the activation function. In ANN, based on the threshold value the activation functions are defined and the output is calculated.

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- **Learning Rate (α)**

- The learning rate is used to control the amount of weight adjustment at each step of training. The learning rate ranges from 0 to 1. It determines the rate of learning at each time step

The notations mentioned in this section have been used in this textbook for explaining each network.

- x_i : Activation of unit X_i , input signal.
- y_j : Activation of unit Y_j , $y_j = f(y_{inj})$
- w_{ij} : Weight on connection from unit X_i to unit Y_j .
- b_j : Bias acting on unit j . Bias has a constant activation of 1.
- W : Weight matrix, $W = \{w_{ij}\}$
- y_{inj} : Net input to unit Y_j given by $y_{inj} = b_j + \sum_i x_i w_{ij}$
- $\|x\|$: Norm of magnitude vector X .
- θ_j : Threshold for activation of neuron Y_j .
- S : Training input vector, $S = (s_1, \dots, s_i, \dots, s_n)$
- T : Training output vector, $T = (t_1, \dots, t_j, \dots, t_n)$
- X : Input vector, $X = (x_1, \dots, x_i, \dots, x_n)$
- Δw_{ij} : Change in weights given by $\Delta w_{ij} = w_{ij}(\text{new}) - w_{ij}(\text{old})$
- α : Learning rate; it controls the amount of weight adjustment at each step of training.
- ρ : Vigilance parameter, it controls the number of clusters in unsupervised networks.

